

Dan Ogawa Lillrank

Robotics Researcher/Software Engineer

Areas of specialization

Robotics Automation (Software)

- Manipulation
 - Inverse Kinematics
 - Trajectory Planning
 - (Deep) IL/RL in robotics
- Simulation:
 - IsaacSim
 - Mujoco (sim2real)
 - Drake
 - pybullet
- Navigation:
 - SLAM
 - Path Planning

Machine Learning

Reinforcement Learning (RL)
Imitation Learning (IL)
Computer Vision(CV)
Deep Learning(DL)
Vision Language Models

Programming

python	6+ years
C++	5 years
Matlab	3 years
C	2 years

Technical

Linux, Pytorch, Llama, scikit-learn, ROS(Moveit!), OpenCV, PCL, CUDA, MPI, Docker, Git, Jira, Confluence, AWS

Languages

English	Fluent
Swedish	Native
Japanese	Native

-  /in/dan-lillrank
-  Harimus
-  Scholar
-  dan.lillrank@gmail.com
-  Stockholm, Sweden
-  (+46)0705211447

WORK EXPERIENCES

2021-2024



Chief Researcher

ARAYA ·  Tokyo, Japan

Neurotechnology R&D Unit, Deep Reinforcement Learning research team.

Designed and implemented a comprehensive robotics stack utilizing ROS and Moveit! and Vision Language model for a Brain Machine Interface project, leading to successful integration of Franka robot arm controlled by human brain signals. Led a team of junior researchers and engineers. Team grew from 2 to 8.

Robotics manipulation, Python, pytorch, ROS, Mujoco, IsaacSim DL, RL, IL, CV, Diffusion models

2021-2022



Technical staff

AIST NATIONAL INSTITUTE OF ADVANCED INDUSTRIAL SCIENCE & TECHNOLOGY ·  Tokyo, Japan

Technical Staff/Research at AIST Automation Research team, working closely with Ogata lab. Utilizing Deep Predictive models for robot manipulation tasks.

PyTorch, Object Segmentation, CUDA, ROS, Gazebo-sim

2020-2021



Robotics Software Engineer

TELEXISTENCE ·  Tokyo, Japan

Main software integrator for the automation team. Implemented Pick & Place workflow for the Model-T humanoid upper-torso robot by integrating perception and planning/control modules using ROS. Integration tested on the custom robot hardware. Integrated several custom grippers to the robot.

Humanoid robot, C/C++, Python, ROS, OpenCV, OpenRave

2019-2020



Robotics Software Engineer

QBIT ROBOTICS ·  Tokyo, Japan

Engineered a robust facial recognition and multi-object tracking system using ROS, Docker, and Python. Proposed & created a pipeline to store the customer information in the cloud. Data-visualization was delivered to the sales team & store manager.

ROS, OpenCV, Object tracking, scikit-learn, panda, AWS

PUBLICATIONS

- 2025 **Improving Low-Cost Teleoperation: Augmenting GELLO with Force**
IEEE/SICE International Symposium on System Integration
- 2024 **A pragmatic look at deep imitation learning.**
Asian Conference on Machine Learning.
- 2024 **A comparison of visual and auditory EEG interfaces for robot multi-stage task control**
Frontiers in Robotics and AI
- 2019 **Registration algorithms for matching laser scans in robotics application**
Thesis work in KTH diva portal

EDUCATION

- 2019 **M.Sc · Systems, Control & Robotics**
 KTH Royal Institute of Technology, Sweden
- 2017 **B.Sc · Engineering Physics**
 KTH Royal Institute of Technology, Swede
- 2017 **M.Sc · Exchange student**
 Kyoto University, Japan